

Rev F custom metal parts - quotation package

Six custom parts only: three lower A6061 adapter plates and three supplier-finished S45C upper PHS6 pocket brackets. No cart, payload, or person. Supervised indoor self-weight PoC only.

Part	Qty	Material	Required process	Authoritative file
A1/A2/A3 LOWER	1 each	A6061-T6/T651	rectangle stock + drilling + M8 tapping	detailed STEP + DXF/hole table
UP-A1/A2/A3	1 each	S45C untreated	supplier-finished 3D milling/drilling	STEP; PDF notes

File priority: upper bracket STEP controls the 3D pocket; upper DXF only documents the mounting face. The detailed lower STEP shows M8 thread geometry, while the lower DXF and CSV hole table control hole locations and tapping callouts. Ask the supplier to flag any unmachinable corner or inaccessible tool path instead of silently changing geometry.

Status: quotation/fabrication candidate. Vendor DFM review and delivered-part fit checks remain. This document does not claim certification, payload use, human carrying, or field safety.

A1 LOWER - LMB radial adapter

A6061-T6 or A6061P-T651 plate, 120 x 70 x 8. Datum is the plate centre. X is the 120 mm direction; Y is the 70 mm direction. No outline cutting beyond the ordered rectangle.

Hole	X	Y	Machining	Function
P1	-48.000	12.000	D9 THRU	profile bolt
P2	48.000	12.000	D9 THRU	profile bolt
T1	14.500	-24.000	drill D6.8, tap M8x1.25 THRU	LMB-10
T2	14.500	12.000	drill D6.8, tap M8x1.25 THRU	LMB-10

Machining notes: deburr all edges and holes; break sharp edges lightly. Keep the two M8 axes normal to the plate. Verify LMB-10 hole pitch against the delivered bracket before tapping. Detailed STEP models M8x1.25 threads. The quote-simplified STEP shows D6.8 pilot holes; this callout controls thread tolerance and class.

Overall	Material	Profile holes	LMB holes	Surface
120 x 70 x 8	A6061-T6/T651	2 x D9 THRU	2 x M8x1.25 THRU	as-machined / deburred

A2 LOWER - LMB radial adapter

A6061-T6 or A6061P-T651 plate, 120 x 70 x 8. Datum is the plate centre. X is the 120 mm direction; Y is the 70 mm direction. No outline cutting beyond the ordered rectangle.

Hole	X	Y	Machining	Function
P1	-48.000	-18.557	D9 THRU	profile bolt
P2	48.000	-18.557	D9 THRU	profile bolt
T1	23.927	12.000	drill D6.8, tap M8x1.25 THRU	LMB-10
T2	-7.250	-6.000	drill D6.8, tap M8x1.25 THRU	LMB-10

Machining notes: deburr all edges and holes; break sharp edges lightly. Keep the two M8 axes normal to the plate. Verify LMB-10 hole pitch against the delivered bracket before tapping. Detailed STEP models M8x1.25 threads. The quote-simplified STEP shows D6.8 pilot holes; this callout controls thread tolerance and class.

Overall	Material	Profile holes	LMB holes	Surface
120 x 70 x 8	A6061-T6/T651	2 x D9 THRU	2 x M8x1.25 THRU	as-machined / deburred

A3 LOWER - LMB radial adapter

A6061-T6 or A6061P-T651 plate, 120 x 70 x 8. Datum is the plate centre. X is the 120 mm direction; Y is the 70 mm direction. No outline cutting beyond the ordered rectangle.

Hole	X	Y	Machining	Function
P1	-50.000	6.557	D9 THRU	profile bolt
P2	50.000	6.557	D9 THRU	profile bolt
T1	-18.427	12.000	drill D6.8, tap M8x1.25 THRU	LMB-10
T2	12.750	-6.000	drill D6.8, tap M8x1.25 THRU	LMB-10

Machining notes: deburr all edges and holes; break sharp edges lightly. Keep the two M8 axes normal to the plate. Verify LMB-10 hole pitch against the delivered bracket before tapping. Detailed STEP models M8x1.25 threads. The quote-simplified STEP shows D6.8 pilot holes; this callout controls thread tolerance and class.

Overall	Material	Profile holes	LMB holes	Surface
120 x 70 x 8	A6061-T6/T651	2 x D9 THRU	2 x M8x1.25 THRU	as-machined / deburred

UP-A1 - PHS6 supplier-finished pocket bracket

S45C untreated, one finished part. Local origin is the PHS6 ball centre. The mounting pad is rotated 90.000 degrees about local Z in the supplied STEP. Quote and machine from STEP; the DXF is mounting-face reference only.

Feature	Nominal dimension	Manufacturing note
Base	60 x 30 x 9	two D6.6 mount holes, pitch 44
PHS6 saddle	D20.2 x 6.75 wide	STEP controls 3D pocket; remove burrs
Saddle outside	D28	support ring joined to two 4 mm rails
Centre fastener	D6.6 THRU	D11 counterbore x 6 deep from top
Side insertion channel	18.5 wide	keep open for PHS6 housing installation
Height	ball centre to mount face: 39	do not change without assembly re-check

Suggested quote notes: general dimensional tolerance +/-0.2 unless the supplier proposes a different DFM tolerance; saddle D20.2 is a clearance feature, not a press fit; no heat treatment or plating; deburr C0.2 max. The supplier may propose tool-radius relief only if the PHS6 contact surface and the D20.2 clearance remain unchanged.

Critical assembly note: fit the PHS6 housing and its central M6x12 low-head retainer before the bracket is bolted to the upper profile. The bearing ball, actuator eye, pin and E-ring are separate parts and must not be clamped by the pocket.

UP-A2 - PHS6 supplier-finished pocket bracket

S45C untreated, one finished part. Local origin is the PHS6 ball centre. The mounting pad is rotated -150.000 degrees about local Z in the supplied STEP. Quote and machine from STEP; the DXF is mounting-face reference only.

Feature	Nominal dimension	Manufacturing note
Base	60 x 30 x 9	two D6.6 mount holes, pitch 44
PHS6 saddle	D20.2 x 6.75 wide	STEP controls 3D pocket; remove burrs
Saddle outside	D28	support ring joined to two 4 mm rails
Centre fastener	D6.6 THRU	D11 counterbore x 6 deep from top
Side insertion channel	18.5 wide	keep open for PHS6 housing installation
Height	ball centre to mount face: 39	do not change without assembly re-check

Suggested quote notes: general dimensional tolerance +/-0.2 unless the supplier proposes a different DFM tolerance; saddle D20.2 is a clearance feature, not a press fit; no heat treatment or plating; deburr C0.2 max. The supplier may propose tool-radius relief only if the PHS6 contact surface and the D20.2 clearance remain unchanged.

Critical assembly note: fit the PHS6 housing and its central M6x12 low-head retainer before the bracket is bolted to the upper profile. The bearing ball, actuator eye, pin and E-ring are separate parts and must not be clamped by the pocket.

UP-A3 - PHS6 supplier-finished pocket bracket

S45C untreated, one finished part. Local origin is the PHS6 ball centre. The mounting pad is rotated -30.000 degrees about local Z in the supplied STEP. Quote and machine from STEP; the DXF is mounting-face reference only.

Feature	Nominal dimension	Manufacturing note
Base	60 x 30 x 9	two D6.6 mount holes, pitch 44
PHS6 saddle	D20.2 x 6.75 wide	STEP controls 3D pocket; remove burrs
Saddle outside	D28	support ring joined to two 4 mm rails
Centre fastener	D6.6 THRU	D11 counterbore x 6 deep from top
Side insertion channel	18.5 wide	keep open for PHS6 housing installation
Height	ball centre to mount face: 39	do not change without assembly re-check

Suggested quote notes: general dimensional tolerance +/-0.2 unless the supplier proposes a different DFM tolerance; saddle D20.2 is a clearance feature, not a press fit; no heat treatment or plating; deburr C0.2 max. The supplier may propose tool-radius relief only if the PHS6 contact surface and the D20.2 clearance remain unchanged.

Critical assembly note: fit the PHS6 housing and its central M6x12 low-head retainer before the bracket is bolted to the upper profile. The bearing ball, actuator eye, pin and E-ring are separate parts and must not be clamped by the pocket.

Assembly interface summary

Interface	Selected stack	Practical installation order
Lower	LMB-10 -> A1/A2/A3 M8 taps -> profile D9 holes	fit plate to profile, then bolt LMB-10
Upper housing	PHS6 -> CBS6-12 -> UP-Ax pocket	install PHS6 and M6 retainer on bench
Upper pin	HCDGH6-35 + WSSB10-6-4 + actuator eye + WSSB10-6-1.5 + PHS6 ball + supplied E-ring	assemble after bracket is on profile
Upper profile	E-DNF3030 + E-SPN306 + E-DCBK3025	preload slot nuts before frame ends close

Representative digital checks are limited to neutral, Z=50 mm, and simultaneous +/-3 degree pitch/roll combinations. They are interference screens, not proof of continuous motion, strength, tolerance stack, or supplier acceptance.

Before motor power: measure the delivered eye width and bore, dry-fit each pin stack, confirm the PHS6 housing seats without clamping the ball, and verify that all three actuators can be moved one axis at a time within the software travel window.